

Department of Computer Science & Engineering (CSE)

MGT 417: Industrial and Operational Management

08/12/2023

Mid Term Examination

Semester: Fall 2023

Time: 1.5 hours

Total Marks: 10 x 3 = 30

Instructions:

- All parts of each question must be answered consecutively.
- The figures in the right margin indicate full marks.
- Answer any **Three (3)** of the following questions.

	Marks
1. A. Briefly explain the primary functions of business organizations.	3
B. Illustrate the operations functions with required example.	2
C. Critically analysis the difference between production of goods and providing service.	5
2. A. Explain the various reasons for which organizations fails.	4
B. Discuss the SWOT analysis with necessary illustration.	6
3. A. Discuss the impact of the four industrial revolutions.	5
B. Critically analysis the key issues for today's business operations.	5
4. A. Describe the elements of a good forecast.	4
B. Briefly explain the reasons of forecast error.	2
C. ABC Co. Ltd. is a manufacturer of ceiling fan.	4

Year	Annual Sales
2019	Tk. 500,000
2020	Tk. 700,000 0.05
2021	TK. 650,000 0.15
2022	Tk. 500,000 0.3
2023	Tk. 550,000 0.5

Requirement: Forecast the sales for 2024 using the weighted average forecasting. The weight is 0.5, 0.3, 0.15 and 0.05 for 2023, 2022, 2021, and 2020 respectively.

- | | | |
|----|---|---|
| 5. | A. Illustrate "Balanced Scorecard" with necessary examples. | 4 |
| | B. XYZ Co. Ltd. is a manufacturer of Trolley Bag. | 6 |

Details	2022	2023
Labor	Tk. 1,000 per unit	Tk. 1,200 per unit
Materials	Tk. 3,000 per unit	Tk. 2,500 per unit
Overhead	Tk. 200,000	Tk. 5,00,000
Output	500 units at Tk. 5,000	1000 units at Tk. 5,000

Requirements:

- i. Calculate the productivity for the year 2022 and 2023.
- ii. Calculate the productivity growth for the year 2023.



HAMDARD UNIVERSITY BANGLADESH

Department of Computer Science and Engineering
Faculty of Science, Engineering and Technology

Course Code: MAT323
Mid-Term Examination
Date: 12/22/2023

Total Time: 1Hours:30 Minutes.

Course Title: Numerical Analysis
Semester: Fall 2023
Total Marks: 30

$3 \times 10 = 30$

Answer any three (03) of the following questions:

1. (a) Explain the operator E , ∇ and Δ [3]
 (b) Find the approximate root of the equation $f(x) = x^3 - x - 1 = 0$ by using bisection method with two decimal places. [3]
 (c) Prove the given relation [4]
 (i) $\Delta - \nabla = \Delta \nabla$.
 (ii) $E \nabla = \nabla E = \Delta$. [2]
2. (a) Write down a short note of false position method. [2]
 (b) Find the approximate root of the equation $f(x) = 3x - \cos x - 1$ by using false-position method. [4]
 (c) Find the approximate root of the equation $f(x) = x \sin x + \cos x$ by using Newton-Raphson method. [4]
3. (a) State the Lagrange's interpolation formula for $n + 1$ data. [2]
 (b) Find the value of the function $y = f(x)$ for $x = 102$ and x for $f(x) = 13.5$ from the following table [using Lagrange's interpolation formula]: [8]

x	93	96.20	100	104.20	108.70
$y = f(x)$	11.38	12.80	14.70	17.07	19.91
4. (a) Derive the Newton's forward formula. [3]
 (b) What is interpolation? Find the value of the function $y = f(x)$ for $x = 7.5, x = 9.5$ from the following table: [7]

x	1	2	3	4	5	6	7	8
$F(x)$	1	8	27	64	125	216	343	512
5. (a) Find the approximate root of the equation $f(x) = e^x - x = 0$ by using Iteration method. [3]
 (b) Find the value of the function $y = f(x)$ for $x = 8, x = 12$ from the following table: [7]

x	4	5	7	10	11	13
$f(x)$	48	100	294	900	1210	2028



Faculty of Science, Engineering and Technology
Department of Computer Science and Engineering
CSE 323 : Operating System

Batch: 20th

Semester: Fall 2023

Date: 15/12/2023

Midterm Examination

Time: 1 Hour 30 Minutes

Total Marks: 30

Answer any **five** questions.

5 × 6 = 30

1. (a) Define Operating System. Mention some of the Usage of Operating System. 3
✓ (b) What are the roles of Operating System? Briefly Explain. 3
2. (a) State the Dual-Mode operation of CPU? 2
(b) What is Kernel? Briefly describe the components of a Kernel. 4
3. (a) Define System Call with an example. 2
(b) What are the steps in making a System Call? 4
4. (a) Distinguish between a Program and a Process. 3
(b) Illustrate how Process is represented in Memory. 3
5. (a) Explain the **lifecycle** of a Process. 3
✓ (b) What is the **Process Control Block**? Mention some of its Components. 3
6. (a) Write short note on **Multithreading**. 2
(b) Explain Context Switching with necessary figure. 4
7. (a) Why does each thread have its own stack? 3
(b) Briefly describe the Hybrid Implementation of Thread. 3

Faculty of Science, Engineering and Technology
Department of Computer Science and Engineering
CSE 327: Computer Peripherals and Interfacing

Batch: 20th

Date: 12/12/2023

Mid-Term Examination

Semester: Fall 2023

Time: 1 hour 30 minutes

Total Marks: 30

Answer any five questions.

Marks: 5 x 6 = 30

1. (a) Draw how a peripheral device is connected with CPU. 2
✓ (b) Derive the necessity of Computer Interfacing. 4
2. (a) Define Computer Peripherals with example. 2
✓ (b) Draw the basic organization of a computer system and explain. 4
3. (a) Write the importance of Computer Peripherals. 2
(b) Briefly explain the interfacing function with interfacing elements. 4
4. (a) Define the Input-Output mode of a Programmable Peripheral Interface. 2
✓ (b) Draw the block diagram and explain the working principle of a Programmable Peripheral Interface. 4
5. (a) Draw the classification chart of a Computer according to application and structure. 2
(b) State the features of analog computer and digital computer. 4
6. Design the pin diagram, timing diagram and explain the working principle of a Keyboard Controller. 6

7. How the following set of instruction stores in computer memory?

$$z = (4x+y)-3+2-6$$

Variable	Address	Values
x	00000100	0000010111
y	00000011	0000101
z	00001001	?

Determine the value of z.

Handwritten calculations:

$$\begin{aligned}
 & (4 \times 23 + 5) - 3 + 2 - 6 \\
 & = 92 + 5 - 3 + 2 - 6 \\
 & = 90 \\
 & 1011010
 \end{aligned}$$



HAMDARD UNIVERSITY BANGLADESH

Faculty of Science, Engineering and Technology
Department of Computer Science and Engineering
CSE 321: Theory of Computing

Batch: 20th

Date: 09/12/2023

Mid Term Examination

Semester: Fall 2023

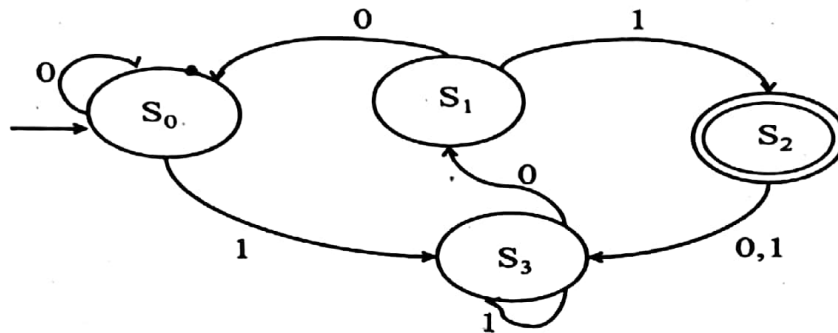
Time: 1 hour 30 minutes

Total Marks: 30

Answer any **five** questions from the following.

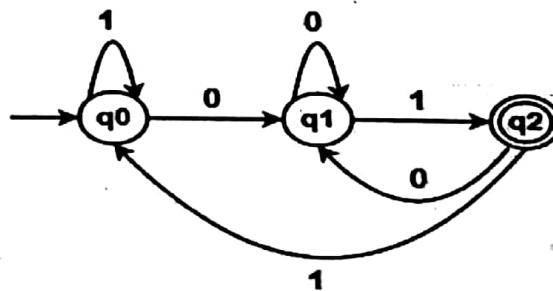
5x6= 30

- 1 (a) Explain the concept of automata theory. 2
(b) Determine the following 4
 - i. Lengths of the following strings: "0010" and "aab".
 - ii. Prefixes of the string "aaabc."
 - iii. Proper suffixes of the string "aaabc."
 - iv. Substrings of the string "aaabc."
- 2 (a) Determine if the string 11001 is accepted or not. 3



- (b) Design an NFA that recognizes strings over $\{0, 1\}$ where '1' is the last character. 3

- 3 (a) Determine the five tuples for the following DFA. 3



- (b) Design a DFA to accept string of 0's and 1's ending with 001. 3

- 4 (a) Design a DFA to accept string that start with 0 and has odd length or start with 1 and has even length. 3

- (b) Design a DFA to accept strings of $\{0, 1\}$ that begin or end with 111. 3

- 5 (a) State the applications of automata theory. 2

- (b) $L = \{\omega | \omega \text{ starts and ends with the same symbol}\}$. Will the string $\omega =$ "abbababba" be accepted? Show the steps using Extended Transition 4

Function.

- 6 (a) What is the difference between Φ and $\{\epsilon\}$? 2
 (b) Design an NFA for the language of all strings over $\{a, b\}$ that contain ababb. 4

- 7 (a) Given an NFA, $N = (\{q_0, q_1, q_2\}, \{0, 1\}, \delta, q_0, \{q_2\})$, with transitions: 3

State	0	1
$\rightarrow q_0$	$\{q_0, q_1\}$	$\{q_1\}$
q_1	$\emptyset$	$\{q_2\}$
$*q_2$	$\{q_1\}$	$\{q_2\}$

Task: Convert this NFA into a DFA.

- (b) Convert the following Mealy Machine to Moore Machine. 3

Current State	Next State			
	0		1	
	State	Output	State	Output
$\rightarrow q_1$	q_3	0	q_{20}	0
q_{20}	q_1	1	q_{40}	0
q_{21}	q_1	1	q_{40}	0
q_3	q_{21}	1	q_1	1
q_{40}	q_{41}	1	q_3	0
q_{41}	q_{41}	1	q_3	0

Faculty of Science, Engineering, and Technology
Department of Computer Science and Engineering
CSF325: System Analysis and Design

Batch: 20th

Semester: Fall 2023

Date: 23 / 12 / 2021

Mid-Term Examination

Time: 1 Hour 30 Minutes

Total Marks: 30

Answer any three questions of the followings.

3 × 10 = 30

1. (a) Who are the typical stakeholders in an information system and what are their roles? 3
(b) Explain entity, associative entity and attributive entity with example. 3
(c) Fabulous Fashions wants to introduce a new computer system for its perishable products warehouse. The costs and benefits are as follows: 4

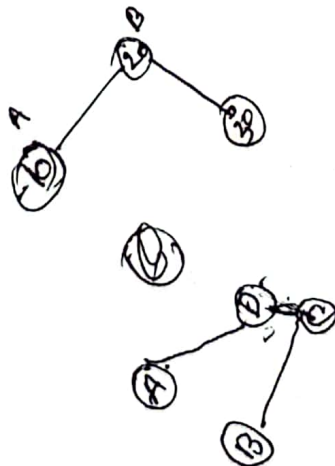
Year	Costs	Benefits
1	\$35,000	\$18,000
2	\$36,700	\$23,500
3	\$34,200	\$25,100
4	\$30,000	\$36,300
5	\$28,800	\$46,100
6	\$26,500	\$68,500

Given a discount rate of 9 percent, perform present value analysis on the data for Alamo Foods and make a comment after the analysis.

2. (a) What are the questions may arise to identify the business objectives? 2
(b) When closed questions are appropriate for use in interviewing? List some advantages and disadvantages of closed interview questions. 4
(c) What should be the concentrate elements of a decision maker's physical environment that a systems analyst can observe by using STROBE? 4
3. (a) Write the guidelines when choosing language for questionnaire. 3
(b) Briefly explain the scaling in questionnaires. What kinds of problem may arise due to careless construction of scales for interview? 3
(c) Why feasibility analysis is required to initiate a project? Describe the key elements of feasibility analysis. 4
4. (a) Explain the reasons for cost estimates fail. 2
(b) Draw a fishbone diagram to identify all the things that can go wrong in developing a system. 4
(c) What is meant by sampling in information gathering? What are the reasons a systems analyst would want to sample data or select representative people to interview? 4

Task	Must Follow	Time (Weeks)	Crash Time	Cost Per Week
A	None	5	3	\$500
B	None	3	2	\$300
C	B	2	2	\$400
D	A, C	3	2	\$750
E	A, C	4	3	\$200
F	D	2	2	\$250
G	E, F	5	3	\$400
H	F	3	2	\$380
I	H	2	2	\$300
J	I	2	2	\$500

- (a) Draw the PERT diagram for the system and identify the critical path. 5
- (b) If Zihad expedites the project as much as he can, calculate the saving amount and the minimum time the project will take. 5



HAMDARD UNIVERSITY BANGLADESH

FACULTY OF BUSINESS ADMINISTRATION

MGT 417 / MGT 419: Industrial and Operations Management

Semester: Fall 2023

06/03/2024

Final Examination

Time: 2.00 Hours

Total Marks: 10 x 4=40

Instructions:

- All parts of each question must be answered sequentially.
- The figures on the right indicate respective marks.
- Answer any four (4) of the following questions.

01. a. Explain the three levels of management decisions and how they impact organizational success. 03
b. Examine your understanding of Productivity by explaining its importance of deriving higher productivity. 03
c. Showcase in drawing the Simple Block Diagram of a Production System along with relevant description. 04
02. a. Draw the Business Model Canvas with an example in it. 06
b. What is Plant Layout? Does the positioning of different elements in the site impact productivity and profitability. 04
03. a. Define Project. What are the features of a Project? 05
b. How would you do the segmentation of a shower gel, produced for women? 05
04. Patients arrive randomly at a clinic to get treated at an average rate of 7 per hour to Dr Rafi. He takes on average 8 minutes for each consultation. Having continuous flow of patients, he is thinking whether he should ask help from a fellow colleague. The arrival process is Poisson, and the service rate can be assumed to be negative exponential. Calculate: 10
i. Total Time in the System or Average waiting time in the Clinic
ii. Average Waiting time in line
iii. Average number of Patients in the System
iv. Average Number of Patients in Line/ Queue Length
v. Utilization Rate of the facility or Probability of the System being Busy

05

05

a.

Activity	Time in Weeks
1-2	7
1-3	6
2-4	8
3-5	9
2-5	6
4-6	4
5-6	5
6-7	8
5-7	11

Find the time to complete the project using Critical Path method.

05

b. Differentiate PERT and CPM.

06

Write Short Notes on:

TQM, JIT, Quality Circle, Kaizen, Supply Chain Management

10

Faculty of Science, Engineering and Technology
Department of Computer Science and Engineering
CSE 327: Computer Peripherals and Interfacing

Batch: 20th

Semester: Fall 2023

Date: 12/03/2024

Final Examination

Time: 2 hours

Total Marks: 40

Answer any five questions.

5 x 8 = 40

- | | | |
|----|---|---|
| 1. | (a) Derive the different types of error check during serial Input-Output transmission. | 4 |
| | (b) Briefly explain the basic requirements of Serial Input-Output of Computer Peripherals. | 4 |
| 2. | (a) Draw the classification chart of computer peripherals. | 3 |
| | (b) Explain the processor to memory communication during information transmission. | 5 |
| 3. | (a) State and explain the physical connection to Micro-processor and internal logic with diagram. | 4 |
| | (b) Briefly explain the read access time and write access time of a RAM. | 4 |
| 4. | (a) State the principles of handshaking method. | 3 |
| | (b) Draw the diagram for CPU connection to I/O devices and CPU with I/O Processor. | 5 |
| 5. | (a) State the transmission mode of serial-interface controller. | 3 |
| | (b) Explain the architecture of serial-interface controller with diagram. | 5 |
| 6. | (a) Briefly explain ARM processor bus interface with diagram. | 3 |
| | (b) Draw the block diagram of ARM core AMBA interface in details. | 5 |
| 7. | (a) Compare between SIMD array and systolic array. | 2 |
| | (b) State and explain the features of systolic arrays. | 3 |
| | (c) Define the types of systolic arrays with diagram. | 3 |



HAMDARD UNIVERSITY BANGLADESH

Faculty of Science, Engineering and Technology
Department of Computer Science and Engineering
CSE 321: Theory of Computing

Batch: 20th

Date: 11/03/2024

Final Examination

Semester: Fall 2023

Time: 2 hours

Total Marks: 40

Answer any five questions from the followings:

5x8= 40

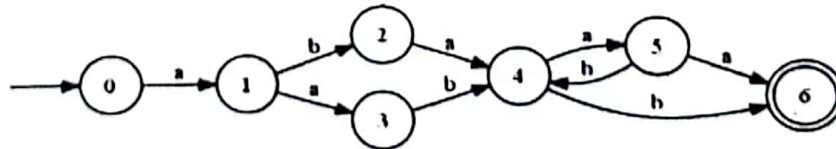
- 1 (a) Outline the tuples associated with the Mealy Machine. 2
✓ (b) Find the Mealy Machine equivalent to the following Moore Machine: 6

Present state	Next state		Output
	0	1	
→A	A	B	0
B	B	C	1
C	C	D	0
D	D	A	0

- 2 (a) Describe the procedure for transforming a DFA to a regular expression using State Elimination. 3
✓ (b) Construct a CFG for a language $L = \{wcwR \mid \text{where } w \in (a, b)^*\}$ and derive a string "abbcbbba". 5

- 3 (a) Provide formal definition of a Context Free Grammar. 2
✓ (b) Derive the string "aabab" for leftmost derivation and rightmost derivation using a CFG given by, 6
 $S \rightarrow AB \mid BA$
 $A \rightarrow a \mid aS \mid bAA$
 $B \rightarrow b \mid aB \mid aSA$

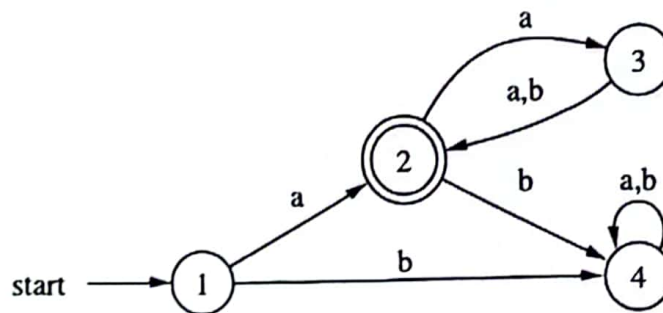
- 4 (a) Define Grammar. 1
✓ (b) Write the RE for the following automata using state elimination. 2



- (c) Show the CYK Algorithm with the following CNF grammar G: 5
 $S \rightarrow AB \mid BC$
 $A \rightarrow BA \mid a$
 $B \rightarrow CC \mid b$
 $C \rightarrow AB \mid a$
 w is ababa
 Is ababa in $L(G)$?

- 5 (a) State the properties of a parse tree. 2
- (b) Show the derivation tree for string "aabbbb" with the following grammar: 2
- $$S \rightarrow AB \mid \epsilon$$
- $$A \rightarrow aB$$
- $$B \rightarrow Sb$$
- (c) Convert the given CFG to CNF. Consider the following grammar G1: 4
- $$S \rightarrow a \mid aA \mid B$$
- $$A \rightarrow aBB \mid \epsilon$$
- $$B \rightarrow Aa \mid b$$

- 6 (a) Explain the concept of duality. 2
- (b) Derive RE for the following FA. 3



- (c) Convert the RE $(ab + a)^*$ to an ϵ -NFA. 3
- 7 (a) How do you define a regular expression? 2
- (b) Convert the following regular expressions to their corresponding language descriptions: 3
- $L((a+b)^*c)$
 - $L(0^*1)$
 - $L((0+1)^*0(0+1)^*)$
- (c) Convert the following language descriptions to their corresponding regular expressions: 3
- All strings of 0's and 1's containing the substring '101'.
 - All string of 0's and 1's beginning with 0 and ending with 1.
 - All strings of 0's and 1's where every '0' is immediately followed by a '1'.



HAMDARD UNIVERSITY BANGLADESH

Faculty of Science, Engineering and Technology
Department of Computer Science and Engineering
CSE 323 : Operating System

Batch: 20th

Semester: Fall 2023

Date: 16/03/2024

Final Examination

Time: 2 Hours

Total Marks: 40

Answer any five questions.

5 × 8 = 40

1. (a) What is Inter Process Communication? 2
(b) Explain the fundamental models of Inter Process communication with necessary figures. 6

2. (a) Define the followings: 4x1=4
i. Race Condition
ii. Critical Region
iii. Mutual Exclusion
iv. Busy Waiting

- (b) State the requirements for the solution to race condition in os. 4

3. (a) Mention the possible solutions to mutual exclusion with busy waiting. 2
(b) Explain the Peterson's solution to mutual exclusion. 6

4. (a) What is the Priority Inversion problem? 2
(b) Consider the following table for 5 jobs arriving at time zero in the given order: 6

Job	Burst Time
1	10
2	29
3	3
4	7
5	12

Now find out which algorithm among FCFS, SJF and Round-Robin with quantum 10, would give the minimum average time.

5. (a) Define Deadlocks in OS. State the necessary conditions for deadlocks. 4
(b) Write the algorithm for deadlock detection with one resource of each type. 4
6. (a) What are the data structures needed for deadlock detection with multiple resource of each type? 3
(b) Briefly explain the methods available for recovery from deadlock. 5
7. (a) Define Semaphore. Mention and describe the operations of semaphore. 4
(b) Explain how you can achieve mutual exclusion and synchronization using Semaphore. 4



HAMDARD UNIVERSITY BANGLADESH

Department of Computer Science & Engineering
Faculty of Science, Engineering and Technology

Course Code: MAT323
Final Examination
Date: 3/19/2024

Total Time: 2HR

Course Title: Numerical Analysis
Semester: Fall 2023
Total Marks: 40

Answer any four (04) of the following questions:

$4 \times 10 = 40$

1. (a) Using Euler's method, solve $\frac{dy}{dx} = x + y^2$; $y(0) = 2$. Find $y(0.4)$, taking $h = 0.1$. [3]
(b) Using second and fourth order Runge-Kutta method, find the value of $y(0.4)$ the following first order differential equation $\frac{dy}{dx} = xy + y^2$; $y(0) = 2$ and compare both methods. [7]
2. Find the approximate value of y for $x = 0.2$ by using fourth order Runge-Kutta method where $\frac{d^2y}{dx^2} + x \left(\frac{dy}{dx}\right)^2 + y^2 = 0$; $y(0) = 1$; $\frac{dy(0)}{dx} = 0$, taking $h = 0.1$. [10]
3. Find the approximate value of $\int_1^2 \frac{dx}{1+x^2}$ by trapezoidal rule, Simpson's $\frac{1}{3}$ rule and also compare your result with exact solution $[I = \tan^{-1} 2 - \tan^{-1} 1]$. [10]
4. (a) Write the procedure of fixed point method for nonlinear system of equations. [3]
(b) Solve the system of non-linear equations by fixed point method. [7]
$$\begin{aligned}x^2 + y &= 4 \\ y^2 + x &= 7\end{aligned}$$
5. (a) Find the approximate root of the following system of non-linear problem by using Newton-Raphson method: [10]
$$\begin{aligned}x^2 - y^2 &= 4 \\ x^2 + y^2 &= 8\end{aligned}$$
6. Find the approximate solution of the following system of equations by using Jacobi and Gause-Seidel method: [10]
$$\begin{aligned}2x + 4y + 15z &= 54.8 \\ x + 12y + 3z &= 39.6 \\ 10x + y - 2z &= 7.74\end{aligned}$$



HAMDARD UNIVERSITY BANGLADESH

Faculty of Science, Engineering and Technology
Department of Computer Science and Engineering
CSE325: System Analysis and Design

Batch: 20th

Semester: Fall 2023

Date: 23 / 03 / 2024

Final Examination

Time: 2 Hours

Total Marks: 40

Answer any five questions of the followings.

5 × 8 = 40

1. (a) Write the rules for constructing a data flow diagram. 3
(b) Draw a data flow diagram for payroll system. 5
2. (a) Differentiate between logical and physical data flow diagrams. 4
(b) Explain the reasons for partitioning data flow diagram. 4
3. (a) What are the ways to represent process logic by using structured English? 4
(b) Suppose a store wanted to illustrate its policy on noncash customer purchases. There are three conditions (sale under \$50, pays by check, and uses credit cards) has only two alternatives. The two alternatives are Y (yes, it is true) or N (no, it is not true). Four actions are possible:
i. Complete the sale after verifying the signature.
ii. Complete the sale. No signature needed.
iii. Call the supervisor for approval.
iv. Communicate electronically with the bank for credit card authorization.
Rules are the combinations of the condition alternatives that precipitate an action. Four rules are:
i. IF N (the total sale is NOT under \$50.00)
ii. IF Y (the customer paid by check and had two forms of ID)
iii. IF N (the customer did not use a credit card)
iv. DO X (CALL THE SUPERVISOR FOR APPROVAL)
Now, generate the decision table for illustrating a store's policy of customer checkout with four sets of rules and four possible actions.
4. (a) Write the guidelines for developing a Prototype. 2
(b) Define agile modeling. State the basic principles of agile modeling. 6
5. (a) What is activity diagram? When it is required? 3
(b) Define use case modeling. Draw a use case diagram for student enrollment. 5
6. (a) What are the objectives of creating an effective database for system design? 3
(b) Define Entity Relationship (ER) diagram. Draw an entity-relationship diagram for patient treatment. 5
7. (a) Describe the anomalies may occur when creating database table. 4
(b) What are the factors to consider when choosing output technology? 4